Learning Objective 9.5: I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, medium 1 is $z < 0$, the plane of incidence is the xz plane, both materials are lossless and nonmagnetic, and $\eta_0 = 376.7\,\Omega$. Use Lesson 40's TM convention: the reflected electric field's tangential reference direction is the same as the incident field's.

1. **The TM coefficients.** A 10 GHz TM wave in air strikes a dielectric with $\epsilon_r = 4$ at $\theta_i = 60^\circ$, so $\theta_t = 25.7^\circ$ (Lesson 39 practice, Question 1).

   (a) Starting from $\widetilde{\mathbf{H}}_i = \hat{\mathbf{y}}\,(E_0/\eta_1)\,e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$, $\widetilde{\mathbf{H}}_r = -\hat{\mathbf{y}}\,(\Gamma_{\mathsf{TM}} E_0/\eta_1)\,e^{-j\beta_1(x\sin\theta_i - z\cos\theta_i)}$, and $\widetilde{\mathbf{H}}_t = \hat{\mathbf{y}}\,(\tau_{\mathsf{TM}} E_0/\eta_2)\,e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$, and the electric fields of Lesson 40, apply the tangential boundary conditions and derive Γ_{TM} and τ_{TM}.

   (b) Evaluate Γ_{TM} and τ_{TM}, and check that $\tau_{\mathsf{TM}} = (1 + \Gamma_{\mathsf{TM}})\cos\theta_i / \cos\theta_t$.

   Γ_{TM} =

   τ_{TM} =

   (c) Compare the reflected power fraction with the TE wave's 32.0% at the same angle, and explain why the TM wave reflects so little here.

   reflected =

2. **The Brewster angle.** Sunlight in air reflects from a flat sheet of glass with $n = 1.50$.

   (a) Show that $\Gamma_{\mathsf{TM}} = 0$ when $\tan\theta_i = n_2/n_1$, and that at this angle $\theta_t = 90^\circ - \theta_i$.

   (b) Find the Brewster angle and the fraction of the TE power that reflects at that angle.

   θ_B =

   TE reflected =

   (c) Describe the polarization of the light reflected at 56.3°, and explain why a polarizing filter can block it.

3. **Total internal reflection.** Light at 633 nm travels in glass, $n_1 = 1.50$, toward a flat glass-air boundary.

   (a) Find the critical angle.

   θ_c =

   (b) At $\theta_i = 60^\circ$, find $\cos\theta_t$, and show that $|\Gamma_{\mathsf{TE}}| = 1$. Find the phase of Γ_{TE}.

   (c) Find the distance into the air over which the evanescent field falls to $1/e$ of its value at the boundary.

   $1/\alpha$ =

   (d) Light in air strikes the glass from outside at any angle. Explain why it never undergoes total internal reflection.

Documentation: _____